Months 16 to 18: MLOps

Quick reference · Senior Data Scientist roadmap · The model registry, a served API, containers, Azure, monitoring and rollback.

B27 Serving and monitoring models

B27-L1 The model registry, versions and aliases

Register every candidate as a version, serve through an alias, resolve the alias once, and keep old versions as rollback targets.

- **Registering a model (MLflow 3.17)**: mlflow.register_model(model_uri, name) creates version 1, 2, ... of a named model.
- **Alias**: A movable name for one version. set_registered_model_alias(name, alias, version); load models:/name@alias.
- **Promote and roll back**: Both move the alias. Keep old versions as rollback targets.
- **Resolve once**: get_model_version_by_alias, then load models:/name/version, so the model and the reported version agree.

B27-L2 Serving a model with FastAPI

Let pydantic schemas validate requests, load the model once in a lifespan function inside an app factory, and test with TestClient as a context manager.

- **Request schema (FastAPI 0.162, pydantic 2.14)**: A pydantic model with Field rules. Invalid input gets 422 with a detail list before your function runs.
- **Loading the model**: Once, in a lifespan function that stores it on app.state. Not per request, not at import time.
- **Status codes**: 503 from /health until the model loads, 422 for invalid input, 413 for too many texts.
- **TestClient**: with TestClient(app) as client: runs the lifespan. Without with, start-up never runs.

B27-L3 Packaging a model service as a container image

Pin serving dependencies, install them before copying code, bind to 0.0.0.0, run as a non-root user, and read settings from the environment at start-up.

- **Pinned serving requirements**: name==version for each package the service needs, and nothing else. The model's scikit-learn version must match.
- **Dockerfile order**: FROM, copy requirements, install, copy code and model, non-root USER, CMD. Docker reuses steps until an input changes.
- **--host 0.0.0.0**: Uvicorn 0.54 defaults to 127.0.0.1, which only the container itself can reach.
- **Settings and secrets**: Read settings from environment variables and check them at start-up. Secrets are injected at run time, never built in.

B27-L4 Deploying the container to Azure Container Apps

Deploy with az containerapp up, keep secrets in Key Vault references passed as secretref variables, probe /health over HTTP, and choose min replicas for your latency needs.

- **az containerapp up --source .**: Creates the resource group, environment, Log Analytics and a registry, builds from the Dockerfile, deploys and prints the URL.
- **Secrets in Container Apps**: --secrets name=value or a Key Vault reference; --env-vars NAME=secretref:name. A secret change needs a restart or a new revision.
- **Probes**: Startup, liveness, readiness. HTTP passes on 200 to 399. The default with ingress is TCP, which passes before the model loads.
- **Default scaling**: HTTP rule, 0 to 10 replicas, 10 concurrent requests each. Scale to zero costs nothing but adds a cold start.

B27-L5 Monitoring a served model for drift

Log structured records without raw text, compare inputs and outputs with a fixed-bin PSI, treat 0.1 and 0.25 as starting points, and confirm drift with labels.

- **What to monitor without labels**: Input distributions, predicted class shares, confidence, error rate and p95 latency.
- **PSI**: sum((a - e) * ln(a / e)) over the reference's quantile bins, with shares floored at a small value.
- **PSI bands**: Below 0.1 stable, 0.1 to 0.25 moderate, above 0.25 large: a rule of thumb from credit scoring. Tune it to your data.
- **Prediction logs**: One structured record per request with version, counts, confidence and latency. Never the respondent's text.

B27-L6 Retraining, promotion and rollback

Compare champion and challenger on the same recent data, shadow before switching, promote by moving an alias and deploying a new revision, and keep old versions, images and revisions for rollback.

- **Champion and challenger**: Compare on the same recent labelled batches. Promote only for a gain above a margin, won on most batches.
- **Shadow mode**: Serve the champion, run the challenger on the same requests, log both, and review agreement and swaps.
- **Promotion**: Move the alias, export, build an image tagged with the version, deploy a new revision.
- **Rollback**: Move the alias back and return traffic to the old revision, or redeploy the old image tag. Keep old versions, tags and revisions.